CKSci Grade 4 Science Interactive Notebook

Chapter 1 Printable Inserts: Energy Causes Change & Stored Energy

Welcome Energy Detectives! This printable packet contains flat paste-in inserts, visual vocabulary frames, before/after model boxes, and 1 high-value lift-the-flap foldable for your composition notebook. Follow the teacher's slides to complete your **WRITE**, **DRAW**, and **THINK** tasks!

Insert #	Notebook Page Topic	Format & Type	Estimated Assembly
Insert 1	Visual Vocabulary (Energy, Change, Evidence)	Flat Paste-In (Fill & Sketch)	2 mins
Insert 2	Core Anchor: ENERGY → CHANGE → EVIDENCE	Flat Diagram & Before/After Boxes	2 mins
Insert 3	Forms of Energy Memory Sketchbook	Flat 5-Box Grid (Drawing Clues)	2 mins
Insert 4	Stored Energy: Ready for Change	1 High-Value Lift-the-Flap Foldable	4 mins
Insert 5	Lab Log: AP 1.1 Energy Scavenger Hunt	NOTICE → EVIDENCE → EXPLAIN Log	2 mins
Insert 6	Chapter 1 Reflection & CER Evidence Tracker	Flat Claim-Evidence-Reasoning Box	2 mins

Teacher Pacing & Assembly Guidelines:

- **Flat Paste-Ins First:** Trim along dashed lines and glue flat into composition notebook.
- **Drawing-Heavy Focus:** Definitions use simple fill-in sentence frames. Sketches are visual memory clues.
- **One High-Value Foldable:** Insert 4 features a 3-tab Lift-the-Flap foldable for Stored Energy. Cut dashed lines, fold solid line.
- **Slide Alignment:** Show completed visual models on slides while encouraging students' unique drawing ideas.

Insert 1: Visual Vocabulary (Chapter 1 Anchor)

Glue flat onto Page 2 of your notebook. Fill in the blanks and sketch a memory clue for each term.

ENERGY **WRITE:** Energy is the ability to cause _____.	**DRAW YOUR MEMORY CLUE:** Draw a picture showing energy causing a change: [Draw flashlight turning on or ball rolling]
CHANGE **WRITE:** Change means to become _____.	**DRAW YOUR MEMORY CLUE:** Draw a Before & After simple change: Before: _____ → After: _____
EVIDENCE **WRITE:** Evidence is proof that can be _____.	**DRAW YOUR MEMORY CLUE:** Draw something you can OBSERVE (see, hear, feel): [Draw glowing light or ringing bell]
STORED ENERGY **WRITE:** Stored energy is energy that can cause change _____.	**DRAW YOUR MEMORY CLUE:** Draw one object with stored energy: [Draw sitting battery, stretched band, or raised ball]

Insert 2: Core Unit Model — ENERGY → CHANGE → EVIDENCE

Glue flat onto Page 3 of your notebook. Trace the arrows and complete the observation boxes.

Practice Investigation Examples: Fill in the evidence box for each phenomenon.

Phenomenon	Before State	After State	Observable Evidence (What changed?)
Flashlight	Flashlight is dark / off.	Switch is clicked ON.	The flashlight changed from _____ to _____.
Toy Car	Car is sitting still.	Car is pushed.	The car changed from _____ to _____.
Desk Bell	Bell is quiet.	Bell is struck.	The bell changed from _____ to _____.

ENERGY DETECTIVE THINKING RULE:

- **OBSERVATION:** What did I actually see, hear, or feel? *(Example: 'The light appeared.')*
- **EXPLANATION:** What does my observation mean? *(Example: 'That change is evidence that energy was involved.')*

Insert 3: Forms of Energy Memory Sketchbook

Glue flat onto Page 4 of your notebook. Draw a quick memory clue sketch for each form of energy.

MOTION ENERGY Energy in moving objects	**DRAW YOUR MEMORY CLUE:** [Sketch a rolling ball, running shoes, or spinning fan]
LIGHT ENERGY Energy that we can see	**DRAW YOUR MEMORY CLUE:** [Sketch a glowing bulb, flashlight beam, or sun]
SOUND ENERGY Energy we can hear (vibrations)	**DRAW YOUR MEMORY CLUE:** [Sketch a ringing bell, guitar string, or drum]
HEAT (THERMAL) Energy we feel as warmth	**DRAW YOUR MEMORY CLUE:** [Sketch a warm cup, campfire, or radiator]
ELECTRICAL ENERGY Energy delivered through wires	**DRAW YOUR MEMORY CLUE:** [Sketch an electrical plug, wire, or battery]

Insert 4: High-Value Foldable — Stored Energy: Ready for Change

✂ Cut around outer dashed border. Cut along vertical dashed lines to create 3 flaps. Fold back along top solid line. Glue BASE into notebook Page 5.

STORED ENERGY: READY FOR CHANGE

Stored energy is energy that has the ability to cause a change LATER!

TOP FLAPS (Cut vertical dashed lines between items, fold along solid line):

FLAP 1: BATTERY	FLAP 2: RUBBER BAND	FLAP 3: RAISED OBJECT
[Draw battery on table]	**[Draw stretched band]**	**[Draw ball held high]**
Front: Still on table	***Front: Stretched still***	***Front: Held above ground***

UNDERNEATH THE FLAP (Glue this base page flat into your notebook):

FUTURE CHANGE 1:	FUTURE CHANGE 2:	FUTURE CHANGE 3:
Put into flashlight & switch ON. **Evidence:** Light appears! *Sentence frame:* It causes change later by _____.	Release from fingers. **Evidence:** Band snaps & flies! *Sentence frame:* It causes change later by _____.	Let go of object. **Evidence:** Object falls & hits floor! *Sentence frame:* It causes change later by _____.

Insert 5: Lab Log — AP 1.1 Energy Scavenger Hunt

Glue flat onto Page 6 of your notebook. Record evidence from your classroom Energy Detectives mission.

Step / Routine	Scavenger Hunt Findings & Evidence
1. NOTICE (Choose an object)	Object Name: _____ Before State: _____
2. EVIDENCE (Observe the change)	What Action Happened?: _____ Observable Change: _____
3. EXPLAIN (Form of energy)	Form of Energy Involved: [ ] Light [ ] Sound [ ] Motion [ ] Heat [ ] Electrical Explanation: This change is evidence that _____.
4. CONNECT (Core Model)	Write the model chain for your item: **ENERGY** (_____) ➔ **CHANGE** (_____) ➔ **EVIDENCE** (_____)

Scavenger Hunt Item #2:

Step / Routine	Scavenger Hunt Findings & Evidence
1. NOTICE (Choose an object)	Object Name: _____ Before State: _____
2. EVIDENCE (Observe the change)	What Action Happened?: _____ Observable Change: _____
3. EXPLAIN (Form of energy)	Form of Energy Involved: [ ] Light [ ] Sound [ ] Motion [ ] Heat [ ] Electrical Explanation: This change is evidence that _____.
4. CONNECT (Core Model)	Write the model chain for your item: **ENERGY** (_____) ➔ **CHANGE** (_____) ➔ **EVIDENCE** (_____)

Insert 6: Chapter 1 Reflection & CER Evidence Tracker

Glue flat onto Page 7 of your notebook. Complete your Claim-Evidence-Reasoning reflection.

CLAIM (Answer the mystery)	**Question:** How do scientists know energy is present even if we cannot see energy itself? **My Claim:** We know energy is present because energy causes _____, which we can observe as _____.
EVIDENCE (Observations from lab & reader)	Provide 2 specific observations that prove energy was involved: 1. **Observation 1:** _____ _____ 2. **Observation 2:** _____ _____
REASONING (Connect evidence to claim)	Explain WHY your evidence supports your claim: Energy is the ability to cause change. When I observe an object change from _____ to _____, that observable change serves as scientific evidence that energy was involved!

OFFICIAL ENERGY DETECTIVE CERTIFICATE OF MASTERY

I can identify observable change as evidence of energy! Signed: _____ Date: _____